

Assignment 3 - Solved Questions and Answers

Q1

Create a 1D NumPy array of integers from 1 to 20 using np.arange().

Code:

```
import numpy as np
arr = np.arange(1, 21)
print(arr)
```

Output:

```
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
```

Q2

Create a 4×5 array of zeros and ones, then add them.

Code:

```
zeros = np.zeros((4,5))
ones = np.ones((4,5))
result = zeros + ones
print(result)
```

Output:

A 4×5 array containing all 1s.

Q3

Create 8 evenly spaced values between 0 and 1.

Code:

```
arr = np.linspace(0, 1, 8)
print(arr)
```

Output:

```
[0. 0.14285714 0.28571429 0.42857143
 0.57142857 0.71428571 0.85714286 1. ]
```

Q4

Create np.arange(1,25) and reshape to 4×6.

Code:

```
arr = np.arange(1,25).reshape(4,6)
print(arr.shape)
print(arr.ndim)
print(arr.dtype)
```

Answer:

Shape = (4,6)

ndim = 2

dtype = int64 (may vary by system)

Q5

Array = [10,20,30,40,50,60,70,80,90,100]

Code:

```
arr = np.array([10,20,30,40,50,60,70,80,90,100])
```

```
a = arr[:5]
```

```
b = arr[-3:]
```

```
c = arr[::2]
```

Answer:

(a) [10 20 30 40 50]

(b) [80 90 100]

(c) [10 30 50 70 90]

Q6

Create a 5×5 array and extract parts.

Code:

```
arr = np.arange(25).reshape(5,5)
```

```
row2 = arr[2]
```

```
col3 = arr[:,3]
```

```
block = arr[3,2:]
```

Answer:

(a) Row 2 = [10 11 12 13 14]

(b) Column 3 = [3 8 13 18 23]

(c) Top-right 3×3 block =

```
[[ 2 3 4]
```

```
 [ 7 8 9]
```

```
 [12 13 14]]
```

Q7

Broadcasting.

Code:

```
a = np.array([1,2,3,4,5])
```

```
add100 = a + 100
```

```
mul3 = a * 3
```

Answer:

```
a+100 = [101 102 103 104 105]
```

```
a*3 = [3 6 9 12 15]
```

Q8

Element-wise operations.

Code:

```
a = np.array([10,20,30])
```

```
b = np.array([1,2,3])
```

Answer:

```
a+b = [11 22 33]
```

```
a-b = [ 9 18 27]
```

```
a*b = [10 40 90]
a/b = [10. 10. 10.]
```

Q9

Scores = [72,85,91,68,74,88,62,95,77,80]

Code:

```
scores = np.array([72,85,91,68,74,88,62,95,77,80])
```

Answer:

Mean = 79.2

Median = 78.5

Std $\approx$ 10.11

Min = 62

Max = 95

Range = 33

Q10

Random integer array.

Code:

```
arr = np.random.randint(1,100,size=(3,4))
```

```
total_sum = np.sum(arr)
```

```
row_sum = np.sum(arr, axis=1)
```

```
col_mean = np.mean(arr, axis=0)
```

Answer:

Values will vary each run because random numbers are generated.

BONUS

Generate 1000 random normal values.

Code:

```
data = np.random.normal(loc=50, scale=10, size=1000)
```

```
print(np.mean(data))
```

```
print(np.std(data))
```

Answer:

The mean should be close to 50 and the standard deviation should be close to 10 due to the properties of a normal distribution.